Empirical Validation of a Mathematical Framework of Housing

Evidence from the United States

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Abstract

This paper presents an empirical framework for validating a multidimensional mathematical model of housing using publicly available United States data. Housing is represented as a latent state-space consisting of structural, financial, ownership, legal, and occupancy components. Rather than developing a new econometric methodology, the paper synthesizes established approaches from panel econometrics, state-space modelling, Markov processes, stock-flow consistent accounting, and machine learning. The proposed framework provides a unified empirical architecture capable of integrating housing prices, construction, mortgage markets, foreclosures, ownership, and occupancy into a single mathematical representation.

The paper ends with "The End"

1 Introduction

Housing plays a central role in modern economies as consumption capital, investment capital, collateral for borrowing, and a source of financial stability or instability. Previous research has examined housing prices, mortgage markets, household finance, banking, and macroeconomic fluctuations largely in isolation. The previous theoretical paper proposed a multidimensional state-space representation of housing. The objective of the present paper is to demonstrate how that framework can be operationalized and validated using United States data.

Rather than deriving new econometric theory, the paper relies on well-established statistical methods and publicly available datasets while emphasizing the mapping between theoretical constructs and observable variables.

2 Empirical State-Space

For region r at time t, define the latent housing state

$$X_{r,t} = (S_{r,t}, F_{r,t}, O_{r,t}, L_{r,t}, Q_{r,t}),$$

where

S structural housing conditions,

F financial conditions,

O ownership composition,

L legal and foreclosure conditions,

Q occupancy conditions.

Observed data satisfy the measurement equation

$$Y_{r,t} = HX_{r,t} + \varepsilon_{r,t},$$

where H is a measurement matrix and $\varepsilon_{r,t}$ denotes measurement error.

The latent state evolves according to

$$X_{r,t+1} = AX_{r,t} + BU_t + \eta_{r,t},$$

where U_t contains macroeconomic variables such as interest rates, inflation, unemployment, and national income.

3 Data

The empirical analysis may be constructed entirely from publicly available United States datasets.

Variable	Representative Data Source
House prices	FHFA House Price Index
Mortgage rates	Freddie Mac / FRED
Building permits	U.S. Census Bureau
Housing starts	U.S. Census Bureau
Median income	U.S. Census Bureau
Unemployment	Bureau of Labor Statistics
Mortgage originations	HMDA
Foreclosures	National Mortgage Database
Population	U.S. Census Bureau
Vacancy rates	American Community Survey

4 Variable Construction

Latent Variable	Observable Proxy
Structural state	Housing starts, permits, housing stock
Financial state	Mortgage balances, delinquency, LTV
Ownership state	Homeownership rate, investor share
Legal state	Foreclosures, distressed sales
Occupancy state	Vacancy rate, rental share

5 Econometric Framework

A benchmark panel specification is

$$\Delta P_{r,t} = \alpha_r + \lambda_t + \beta' Z_{r,t} + \varepsilon_{r,t},$$

where $P_{r,t}$ denotes the house price index and $Z_{r,t}$ contains mortgage conditions, construction activity, unemployment, and income.

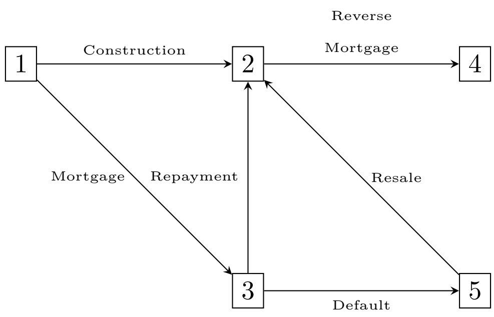
Mortgage default probabilities may be estimated using logistic or hazard models,

$$Pr(D_{r,t} = 1) = \Lambda(\beta' Z_{r,t}),$$

while latent housing dynamics may be estimated using standard state-space methods.

6 Transition Dynamics

The theoretical lifecycle introduced in the previous paper is represented empirically by the transition graph shown below.



Empirical transition probabilities may be estimated from observed frequencies,

$$\hat{p}_{ij} = \frac{\text{Transitions } i \to j}{\text{Observations in state } i},$$

forming an estimated Markov transition matrix.

7 Validation Strategy

The empirical analysis evaluates four hypotheses.

1. The multidimensional state-space is empirically identifiable.
2. Housing transitions exhibit statistically meaningful dynamics.
3. Separating financial, legal, ownership, and occupancy dimensions improves predictive performance.
4. The proposed framework outperforms simpler price-only specifications.

Model performance may be evaluated using RMSE, MAE, AUC, log-likelihood, and out-of-sample forecasting accuracy.

Proposition 1. *The empirical framework nests conventional reduced-form housing regressions as special cases.*

Proof. If the latent state vector is restricted to a single observable component, such as house prices alone, and the remaining dimensions are suppressed, the measurement equation reduces to a standard reduced-form regression model. Consequently, conventional empirical specifications are nested within the proposed multidimensional framework.

□

8 Discussion

The proposed empirical architecture separates theoretical representation from statistical estimation. Any consistent estimator capable of recovering latent states from observed housing variables can be incorporated into the framework. Consequently, the model is compatible with classical econometrics, Bayesian inference, hidden Markov models, Kalman filtering, agent-based simulation, and modern machine learning algorithms.

9 Conclusion

This paper presents a practical empirical strategy for validating a multidimensional mathematical framework of housing using publicly available United States data. By combining established datasets with standard statistical techniques, the framework provides a unified platform for studying housing markets, mortgage finance, ownership structures, foreclosures, and occupancy dynamics. The approach is deliberately modular, allowing future work to incorporate richer spatial detail, micro-level property data, policy evaluation, and AI-based forecasting without altering the underlying mathematical representation.

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Glossary

$X_{r,t}$ Latent regional housing state.

S Structural housing component.

F Financial component.

O Ownership component.

L Legal component.

Q Occupancy component.

H Measurement matrix.

A State transition matrix.

The End